INCH-POUND

MIL-DTL-22263C
11 September 2009
SUPERSEDING
MIL-C-22263B
28 June 1972

DETAIL SPECIFICATION

COUPLING FUEL LINE, FLEXIBLE, 125 PSI GENERAL SPECIFICATION FOR

INACTIVE FOR NEW DESIGN AFTER 18 MARCH 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers flexible couplings for joining variable ferrule type tubing in military aircraft fuel systems.

1.2 Couplings. The coupling materials and sizes are as specified in table I and table II respectively.

TABLE I. Material and code.

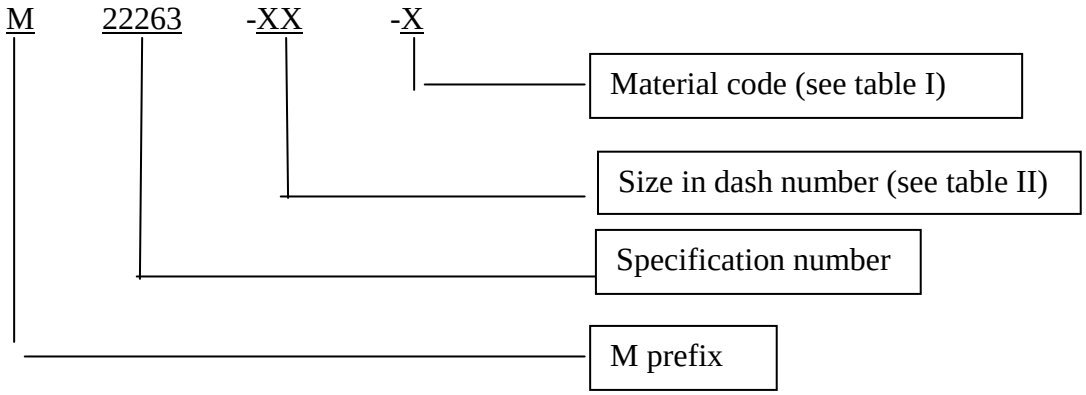
Material	Code
All aluminum	A
Aluminum body and nut, CRES steel retainer	B
All CRES	C
All titanium	T

Comments, suggestions, or questions on this document should be addressed to the Naval Air Systems Command, Attention: Code 4L8000B120-3, Highway 547, Lakehurst, NJ 08733-5100 or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST online database at http://assist.daps.dla.mil.

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1.2.1 Part or identifying number (PIN). The part or identifying number to be used for couplings acquired to this specification is identified as follows:



Example: M22263-10-C = 0.625 tube coupling with CRES material

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-5624	-	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-PRF-7024	-	Calibrated Fluids, Aircraft Fuel System Components
MIL-A-8625	-	Anodic Coatings for Aluminum and Aluminum Alloys
MIL-DTL-25988	-	Rubber, Fluorosilicone Elastomer, Oil and Fuel Resistant, Sheets, Strips, Molded Parts, and Extruded Shapes
MIL-R-25988/1	-	Rubber, Fluorosilicone Elastomer, Oil- and Fuel Resistant, O-Rings, Class 1, Grade 70

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MIL-DTL-83133 - Turbine Fuel, Aviation, Kerosene Type, JP-8 (NATO-F34), NATO F-35 and JP-8+100 (NATO F-37)

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129 - Military Marking for Shipment and Storage

MIL-STD-130 - Identification Marking of US Military Property

MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests

MIL-STD-889 - Dissimilar Metals

MS33784 - Wrenching Features for Cylindrical Nut, Design Standard for (Inactive for new design)

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-470 - Designing and Developing Maintainable Products and Systems, Volume 1

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

SAE INTERNATIONAL

SAE-AMS2700 - Steels, Passivation of Corrosion Resistant (DoD adopted)
SAE-AS8879 - Screw Threads, UNJ Profile, Inch Controlled Radius Root With Increased Minor Diameter (DoD adopted)

(Copies of these documents are available online at http://www.sae.org or from the SAE International, 400 Commonwealth Drive, Warrendale, PA 15096-0001.)

ASTM INTERNATIONAL

ASTM-D3951 - Commercial Packaging (DoD adopted)

(Copies of these documents are available online at http://www.astm.org or from the ASTM International, P.O. Box C700, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.)

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AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NASM20995 - Wire, Safety or Lock (DoD adopted)

NASM33540 - Safety Wiring and Cotter Pinning (DoD adopted)

(Copies of these documents are available online at http://www.aia-aerospace.org or from the Aerospace Industries Association, 1000 Wilson Boulevard, Suite 1700, Arlington, VA 22209-3928.)

ANSI INTERNATIONAL

ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of these documents are available online at http://www.ansi.org or from ANSI, 25 West 43rd Street, New York, NY 10036-8002.)

ASME INTERNATIONAL

ASME B46.1 - Surface Texture (Surface Roughness, Waviness, and Lay)

(Copies of this document are available online at http://www.asme.org or from ASME, 22 Law Drive, P.O Box 2900, Fairfield, NJ 07007-2900.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. The couplings furnished under this specification shall be products that have been inspected and have passed the first article inspection specified herein.

3.2 Materials. Materials shall conform to applicable specifications and shall be as specified herein. Materials that are not covered by applicable specifications or that are not specifically described herein shall be compatible with aviation fuel conforming to MIL-DTL-5624, MIL-PRF-7024 and MIL-DTL-83133 and shall have the lightest practicable weight as possible.

3.2.1 Metals. Metals shall be of a corrosion-resisting type or shall be suitably treated to resist corrosion when exposed to climatic and environmental conditions encountered during the service life of the equipment. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

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3.2.2 Fungus-proof materials. Materials that are not nutrients for fungi shall be used to the greatest extent practicable. In cases where materials that are nutrients for fungi must be used, such materials shall be treated with a fungicidal agent as approved by the procuring activity.

3.3 Finish.

3.3.1 Aluminum alloy parts. Where practicable, all aluminum alloy parts shall be anodized in accordance with MIL-A-8625.

3.3.2 Steel parts. Steel parts shall be corrosion proof or resist to corrosion shall be passivated per SAE-AMS2700. Non-corrosion-resistant steel parts shall not be used.

3.3.3 Titanium parts. Finish shall not be required for titanium parts.

3.4 Dissimilar metals. Connecting tubing with coupling parts shall be compatible and shall meet the requirements of MIL-STD-889.

3.5 Design and construction. The coupling shall be designed for connecting variable ferrule type tubing end when used in aircraft power plant fuel systems. Typical couplings installation shall be as shown in figure 1. The coupling dimensions and weights without O-ring shall be in accordance with table II. The design and construction of the coupling shall be such that it shall withstand the strains, jars, vibrations and other conditions incident to aircraft installations and service usage.

3.5.1 Threads. Machine screw threads shall conform to SAE-AS8879. All external or internal threaded parts shall be securely locked in such a manner as to prevent loosening under test conditions specified herein and under normal service usage. Provisions for safety wiring shall be provided. Safety wires shall be installed in accordance with NASM33540 and shall conform to NASM20995. Wrenching features shall conform to MS33784. Tools shall not be required for torquing.

3.5.2 Angular misalignment. The coupling shall be capable of being installed with a maximum tubing or ferrule installation misalignment of 3 degrees in any direction (see figure 3 and figure 4).

3.5.3 Flexure. The coupling shall provide for one degree flexure in any direction from any installed position. The connector shall maintain a space between tubing and ferrules of 0.062 inch minimum to 0.188 inch maximum.

3.5.4 Permanent deformation. Installation of the coupling on tubing ends under conditions specified herein shall not produce a permanent deformation of the tubing or fitting.

3.5.5 Tubing. The couplings shall be designed for connecting tubing ends conforming to figure 2. The tubing end dimensions shall be in accordance with table III.

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3.6 O-ring seal. The O-ring seal used for testing purposes shall be in accordance with MIL-DTL-25988 of the applicable size specified in table II. The seals shall be easily replaceable without damage to the coupling parts and with a minimum of disturbance or displacement of the installed tubing or fitting. The O-ring seals shall not be subjected to axial loads of tubing under pressure.

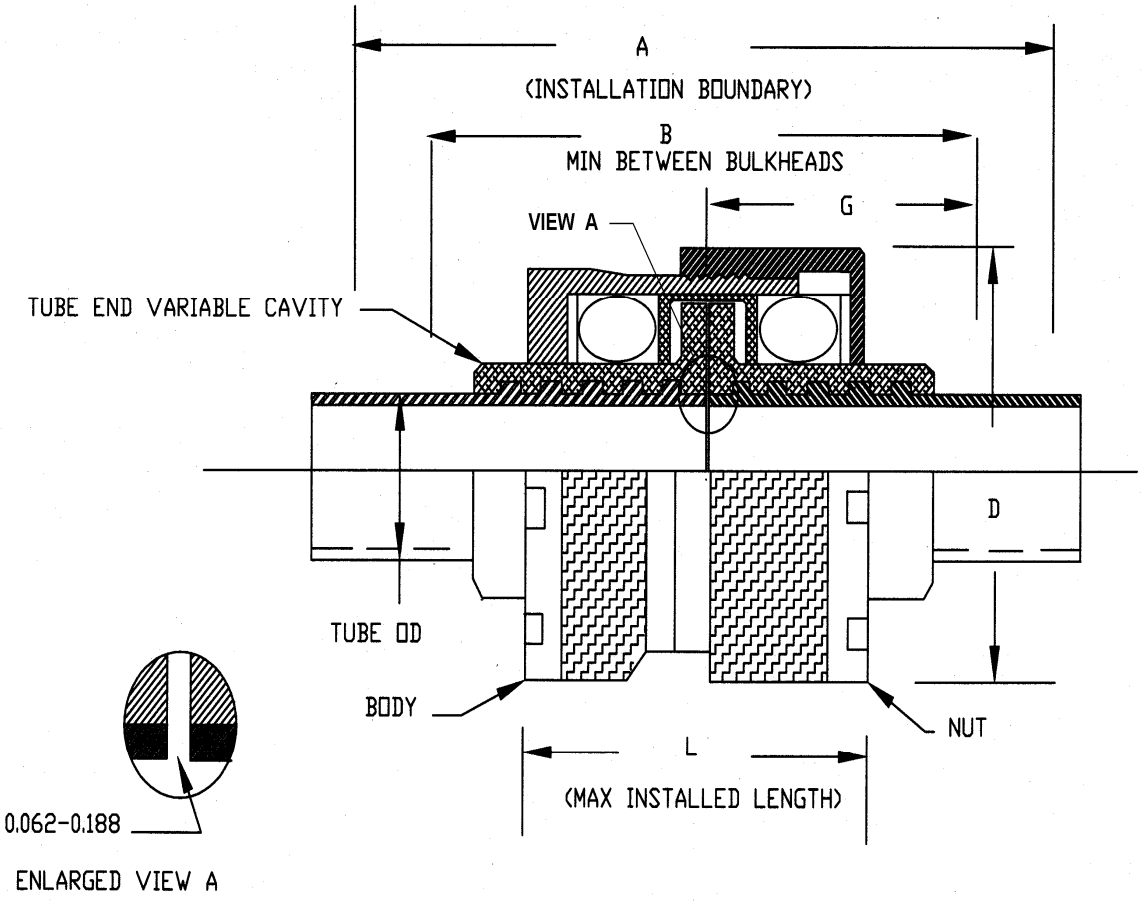


FIGURE 1. Coupling installation connecting two swaged ferrule fuel lines.

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TABLE II. Coupling dimensions and weights. 1/

Dash No	Tube OD	A	B	D (max)	G	L (max)	O-ring Size (Ref) 2/	Weight lb (max)		
								Al	Steel	Ti
-12	0.75	2.51	1.68	1.532	1.31	1.29	-212	0.057	0.163	0.092
-16	1.00	2.57	1.84	1.782	1.37	1.35	-216	0.075	0.215	0.120
-20	1.25	2.57	1.84	2.032			-220	0.086	0.240	0.144
-24	1.50	2.90	2.07	2.407	1.63	1.54	-326	0.128	0.358	0.214
-28	1.75	2.90		2.657			-328	0.176	0.492	0.294
-32	2.00	2.95		3.052	1.65		-330	0.212	0.593	0.354
-40	2.50	2.95		3.427			-334	0.225	0.630	0.376
-48	3.00	2.95		3.927			-338	0.285	0.798	0.476
-56	3.50	3.06	2.17	4.427	1.76	1,63	-342	0.375	1.050	0.627
-64	4.00	3.06	2.18	4.928		1.64	-346	0.399	1.117	0.667
-72	4.50	3.35	2.30	5.678	1.99	1.76	-426	0.654	1.730	0.842
-80	5.00	3.58	2.52	6.178	2.21	1.98	-430	0.762	2.066	1.233
-88	5.50	3.60	2.56	6.553	2.19	2.02	-434	0.777	2.131	1.298
-96	6.00	3.64	2.60	7.053	2.24	2.06	-437	0.932	2.015	1.557

1/ Dimensions are in inches
2/ O-ring sizes conform to MIL-R-25988/1

3.7 Performance. The coupling, when installed in any position (attitude), shall meet all the following performance requirements when tested as specified in section 4 of this specification.

3.7.1 Examination of product. The couplings shall be examined and inspected as specified in 4.7.1 and shall have no defects. All dimensions and weights shall conform to table II.

3.7.2 Proof pressure. The coupling assembly shall be designed to withstand a proof pressure of 250 ± 5 psi. A decrease in pressure when pressurized to proof pressure as specified in 4.7.2 or any leakage from the coupling or joint shall be cause for rejection of the coupling.

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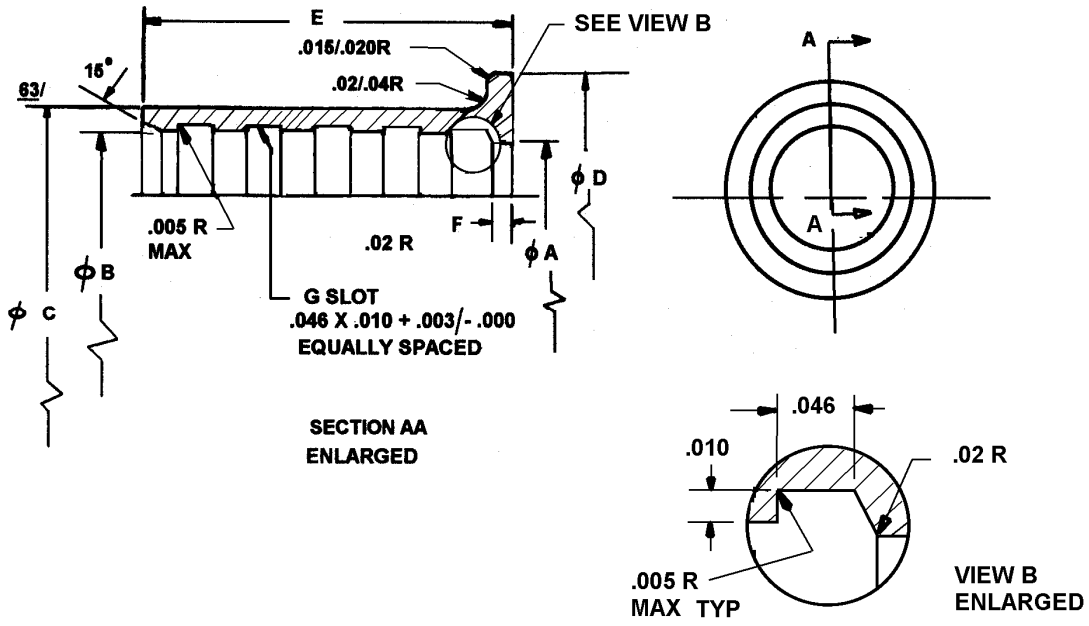


FIGURE 2. Tubing swaged end ferrule dimensions and configuration. 1/

TABLE III. Tubing end, swage end ferrule, variable cavity dimensions. 1/

Dash No	Tube OD	A +.000 -.005	B + .005 - .000	C +.000 -.005	D ± .005	E ± .005	F ± .005	G
-12	0.75	0.730	0.755	0.891	0.984	0.545	0.040	5
-16	1.00	0.980	1.005	1.142	1.265	0.669		
-20	1.25	1.230	1.256	1.392	1.515			
-24	1.50	1.480	1.506	1.642	1.785	0.675		
-28	1.75	1.730	1.756	1.892	2.035		0.055	
-32	2.00	1.980	2.006	2.142	2.308	0.682		
-40	2.50	2.480	2.507	2.642	2.808			
-48	3.00	2.930	3.007	3.145	3.308			
-56	3.50	3.455	3.507	3.650	3.819	0.712		
-64	4.00	3.980	4.008	4.154	4.329			
-72	4.50	4.480	4.508	4.655	4.825			6
-80	5.00	4.980	5.009	5.155	5.300	.956	0.075	
-88	5.50	5.480	5.509	5.660	5.825			
-96	6.00	5.980	6.010	6.165	6.340	1.001		

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1/ NOTES:

1. Materials: 2024 –T4 aluminum, CRES, or Ti
2. Dimensions shown are in inches, unless otherwise specified, tolerance: Decimal ± .005, angle ± 0.5^{o}
3. Unless otherwise specified, break all sharp edges .003 - .015
4. Surface texture in accordance with ASME B46.1

TABLE IV. Tubing sizes and wall thickness.

Coupling sizes	Tube O.D	Wall thickness (min)		
		AL	CRES	Ti
-12	0.75	.035	.020	.020
-16	1.00	.035	.020	.020
-20	1.25	.035	.025	.022
-24	1.50	.035	.025	.022
-28	1.75	.035	.035	.032
-32	2.00	.042	.035	.032
-40	2.50	.042	.035	.032
-48	3.00	.042	.035	.041
-56	3.50	.065	.049	.041
-64	4.00	.065	.049	.041
-72	4.50	.065	.049	.041
-80	5.00	.065	.049	.041
-88	5.50	.065	.049	.041
-96	6.00	.065	.049	.041

3.7.3 Leakage.

3.7.3.1 Vacuum leakage. The coupling assembly shall be subjected to the negative proof pressure specified in 4.7.3.1. A decrease in pressure exceeding 0.5 Hg (0.25 psi) shall be cause for rejection of the test sample.

3.7.3.2 Pneumatic leakage. The coupling shall be tested with pneumatic pressure equal to proof pressure as specified in 4.7.3.2 for 3 minutes at room temperature. Any leakage or evidence of leakage during the test shall be cause for rejection of the test sample.

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3.7.4 Fuel resistance. The coupling assembly shall not leak (sufficient to form a drop) nor show evidence of damage when subjected to high temperature fuel aging at 200 °F, low temperature fuel aging at -65 °F and air dry out at 200 °F when tested as specified in 4.7.4.

3.7.5 Salt fog. The coupling assembly shall pass the salt fog test specified in 4.7.5 without any evidence of excessive corrosion, peeling, blistering of the finish or exposure of base metal. Threads and connecting surfaces and sealing surfaces shall be free of corrosion.

3.7.6 Repeated assembly. There shall be no evidence of deformation, damage or degradation in the connection of the coupling halves after 25 repeated assembly and disassembly test specified in 4.7.6 follow by the proof pressure test without leakage.

3.7.7 Flexure. The coupling assembly, when tested as specified in 4.7.7 with a 3 degree tubing misalignment and flexure at ±0.5 degree for 28,800 cycles followed by the proof pressure test shall show no evidence of leakage, rupture, permanent set, permanent deformation or damage (see figure 4).

3.7.8 Vibration. The coupling assembly shall show no evidence of malfunction or structural failure and shall pass the proof pressure test after exposure to vibration levels as specified in MIL-STD-810 (see table VII) and tested as specified in 4.7.8.

3.7.9 Surge pressure. The coupling assembly shall withstand 50,000 pressure surge cycles without evidence of malfunction or leakage when tested at room temperature specified in 4.7.9.

3.7.10 Burst pressure. The coupling assembly shall be designed to meet the burst pressure requirements at room temperature and extreme temperatures and shall pass the requirements when tested as specified in 4.7.10.

3.8 Maintainability. The coupling assembly when installed in aircraft shall not require periodic maintenance. The seals installed in the coupling shall be easily replaceable without damage to other parts and shall be able to be removed with a minimum of disturbance or displacement of the installed coupling and tubing. The coupling system maintainability requirements shall be integrated with the overall aircraft system maintainability programs and shall comply with the requirements of MIL-HDBK-470.

3.9 Reliability. The coupling assembly shall be designed to withstand the strains, vibrations, and temperature environment incident to aircraft installation and services usage. The reliability of the coupling assembly shall be an integral part of reliability requirements of the total aircraft systems.

3.10 Interchangeability. All parts having the same manufacturer's part number shall be directly and completely interchangeable with each other with respect to installation and performance.

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3.11 Identification of product. Coupling assemblies and parts shall be marked for identification in accordance with MIL-STD-130. The part or identification number (PIN) shall be in accordance with 1.2.1.

3.12 Workmanship. The couplings shall be fabricated and finished with no defects, cracks, burrs and tool marks or sharp edges. Where dimensions are critical and affect the interchangeability of parts, they shall be kept within the tolerances specified in this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.4).

4.2 First article inspection. First article inspection shall consist of all examinations and tests listed in table V.

TABLE V. First article inspection.

Item	Test description	Design requirements	Test requirements
1	Examination of product	3.7.1	4.7.1
2	Proof pressure	3.7.2	4.7.2
3	Leakage	3.7.3	4.7.3
4	Vacuum leakage	3.7.3.1	4.7.3.1
5	Pneumatic leakage	3.7.3.2	4.7.3.2
6	Fuel resistance	3.7.4	4.7.4
7	High temp fuel aging Phase I	3.7.4	4.7.4.1
8	High temp fuel aging Phase II	3.7.4	4.7.4.2
9	Low temp fuel aging	3.7.4	4.7.4.3
10	Air dry out	3.7.4	4.7.4.4
11	Salt fog	3.7.5	4.7.5
12	Repeated assembly	3.7.6	4.7.6
13	Flexure	3.7.7	4.7.7
14	Vibration	3.7.8	4.7.8
15	Surge pressure	3.7.9	4.7.9
16	Burst pressure	3.7.10	4.7.10

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4.2.1 First article sample. Two coupling assemblies of each size shall be subjected to the first article tests by the manufacturer. The tested couplings with untested samples shall be submitted to the procuring activity for final approval. Two complete sets of detail and assembly drawings of the coupling shall be submitted with the samples.

4.2.2 Test assembly. Additional O-ring seals in accordance with 3.6 shall be furnished by the contractor for test purposes. The coupling shall be installed on tubing. The ends shall be mounted so that they are restrained at the maximum angular misalignment specified and in such a manner that they are free to move axially to accommodate end loads due to internal pressure.

4.2.3 Tubing wall thickness. The tubing used in testing the coupling shall have minimum wall thicknesses specified in table IV.

4.2.4 First article data. Upon completion of the first article inspection, the activity responsible for conducting the first article inspection shall forward test data to the contracting activity or acquiring activity with recommendations (see 6.2).

4.3 Test conditions.

4.3.1 Test fluid. Unless otherwise specified, the test fluid shall be in accordance with MIL-PRF-7024.

4.3.1.1 Pressure and temperature. The pressure and temperatures specified in table VI shall apply to all tests required by this specification.

TABLE VI. Temperature and pressure.

Test fluid	Temperature (^{o}F $\pm$ 5 ^{o}F) Fluid and ambient	Pressure ($\pm$ 5 psi)		
		operating	proof	burst
MIL-PRF-7024	-65 ^{o}F to + 200 ^{o}F	125	250	375

4.3.2. Rejection and retest. When a sample from a lot fails the first article tests, acceptance of couplings in the lot shall be withheld until the extent and cause of failure are determined. When corrections have been made, the first article tests shall be repeated.

4.3.3 Resubmitted lot. Where a lot has been rejected by the acquiring activity or contracting activity and before it can be resubmitted for first article test, the cause of rejection and the action taken shall be furnished in writing by the contractor and submitted to the contracting activity or acquiring activity.

4.4 Conformance inspection. Conformance inspection shall consist of:

a. Individual inspection (see 4.5).

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b. Sampling inspection (see 4.6).

4.5 Individual inspection. Each coupling shall be visually examined to determine that the coupling complies with the requirements specified in 4.7.1. Any coupling failing to pass the examination shall be rejected.

4.6 Sampling inspection. The sample size shall be selected in accordance with ANSI/ASQ Z1.4 inspection level I, acceptable quality level (AQL) as specified in the contract and shall pass the tests listed in table VII except to be performed at room temperature. If the sample fails to pass this test, the lot represented by this sample shall be rejected. The tested coupling shall be discarded.

TABLE VII. Sampling inspection.

Test description	Design requirements	Test requirements
Proof pressure	3.7.2	4.7.2
Repeated assembly	3.7.6	4.7.6

4.7 Inspection methods.

4.7.1 Examination of product. The coupling shall be examined to determine conformance to this specification with respect to configuration, dimensions and weight and all the requirements not covered by tests specified herein.

4.7.2. Proof pressure. The test assembly shall be subjected to a proof pressure specified in 3.7.2 for a period of 5 minutes. Any rupture, permanent set, permanent deformation or damage of any part of the test coupling shall be cause for rejection.

4.7.3 Leakage.

4.7.3.1 Vacuum leakage. The test coupling shall be free of fuel, oil and vapor. The coupling shall be connected with vacuum pump with hose or tube having OD 0.5 inch and length 10 ft. A stop valve shall be installed adjacent to the pressure gauge in line with the pump. Pressure shall be measured within 6 inches of the coupling. A negative proof pressure of 28 inches of Hg (14 psi) for sizes -4 through -32 and 12 inches of Hg (6 psi) for sizes -36 through -96 shall be subjected to the coupling and maintain for 15 minutes. A stop valve shall be closed and the pressure shall be monitored for 5 minutes for leakage. The leakage requirements specified in 3.7.3.1 shall be met.

4.7.3.2 Pneumatic leakage. The test coupling shall be thoroughly clean with no external oil or lubricant. The test fluid shall be dry air or nitrogen. The test coupling shall be placed in a protective closure and shall be proof pressure tested at room temperature for 3 minutes. The coupling shall meet the requirement specified in 3.7.3.2.

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4.7.4 Fuel resistance.

4.7.4.1 Phase I high temperature fuel aging. Test fluid shall be circulated through the test assembly at the operating pressure of 125 psi and a temperature of 100°F. The fluid flow shall be in one direction for one hour, and then reversed for a second hour. No indication of flow restriction or blockage of the fluid through the test coupling. The fluid shall then be circulated in one direction through the test assembly at a pressure of 125 psi and a temperature of 160 °F. The fluid temperature shall be maintained at 160 °F for 72 hours. During the final 8 hours of this test, the coupling assembly shall be vibrated in accordance with 4.7.8. Upon completion of this test, the test assembly shall be proof tested at 250 psi at a temperature of 160 °F and held for a minimum of one minute. There shall be no evidence of leakage or deformation of the coupling.

4.7.4.2 Phase II - High temperature fuel aging. Test fluid shall be circulated through the test coupling for 72 hours while at 160 °F and 125 psi. During the final 8 hours of this test, the test coupling shall be vibrated in accordance with 4.7.8. Upon completion, the test assembly shall be proof tested at 250 psi and 160 °F with the pressure to be held for a minimum of one minute. There shall be no evidence of leakage or deformation of the coupling.

4.7.4.3 Low temperature fuel aging. Test fluid shall be circulated through the test coupling for 72 hours while at 125 psi and -65°F. During the final 8 hours of this test, the test coupling shall be vibrated in accordance with 4.7.8. Upon completion, the test assembly shall be proof tested at 250 psi and at -65 °F with the pressure to be held for a minimum of one minute. There shall be no evidence of leakage or deformation of the coupling.

4.7.4.4 Air dry out. Upon completion of the low temperature fuel aging test, the coupling assembly shall be drained and placed in an air oven for 168 hours at 160°F. The low temperature fuel aging test of 4.7.4.3 shall be repeated upon completion of dry out.

4.7.5 Salt fog. Unless otherwise specified in the contract, the coupling assembly shall be exposed to salt fog for 168 hours in accordance with MIL-STD-810, method 509.5. After 168 hours of exposure, the test specimen shall be examined for evidence of corrosion or other damaged of the finish. The proof pressure test shall be performed in accordance with 4.7.2. Any leakage or failure shall be cause for rejection of the test coupling.

4.7.6 Repeated assembly. The coupling assembly shall be tested with repeated assembly and disassembly for 25 operations. There shall be no evidence of damage or degradation in the connecting ability of the coupling. After the repeated assembly test, the coupling assembly shall be subjected to the proof pressure test as specified in 4.7.2. Any leakage or malfunction shall be cause for rejection of the coupling assembly.

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4.7.7 Flexure. The tubing on one side of the coupling shall be rigidly fixed while the other tubing shall be mounted eccentrically on a power driven spindle (see figure 4). The test setup shall provide 0.5° flexure in any direction with an initial 3° misalignment between the centerlines. Test fluid shall be introduced at a pressure of 125 psi. The test fluid temperature shall be 160 °F for the first 4 hour test period and -65 °F during the second 4 hour test period. The test assembly shall be flexed for 8 hours at a frequency of 60 cycles per minute. At the conclusion of this test, a proof pressure test shall be conducted at 250 psi and -65 °F. There shall be no evidence of leakage or deformation of the coupling during any phase of this test.

4.7.8 Vibration test. Unless otherwise specified in the contract, the vibration test shall be conducted in accordance with MIL-STD-810 method 514.6 with the applicable test procedures and test categories as specified in table VIII.

4.7.8.1 Vibration test setup. The test assembly shall be mounted to a vibration table with an angular displacement of 3 degrees between tube center lines (see figure 3). The distance between test assembly table supports shall be 20 inches for all sizes. Locate the test coupling in the center of the table. The major axis of the test assembly shall be perpendicular to the direction of vibration and parallel to the vibration table. The test coupling shall meet the requirements of 3.7.8.

TABLE VIII. Vibration tests.

Type of vehicle	MIL-STD-810		
	Test procedure	Test category	Vibration exposure
Jet aircraft	I	12	Figure 514.6D-1
Propeller aircraft	I	13	Figure 514.6D-2
Helicopter	I	14	Figure 514.6D-3
Ground vehicles	I	20	Figure 514.6D-8
Marine vehicles	I	21	Figure 514.6D-9

4.7.9 Surge pressure.

4.7.9.1 Test setup. Two 2-inch couplings of 2" O.D. and .035" wall thickness shall be assembled with 20 inches lengths of type 6061-T4 aluminum tubing to make up a surge cycle pressure test assembly (see figure 5). Two assemblies shall be used as shown on figure 6. The middle section of each multiple tube assembly shall be in the form of a capped or plugged tee. One end of each assembly shall be capped and ported to provide for filling and pressurization. The other end shall be capped and ported to accept a pressure transducer with a response in the order of 30 milliseconds. Each assembly shall be filled with test fluid and mounted so that movement of tube assemblies shall not be restricted.

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4.7.9.2 Pressurization source. The pressurization source shall be designed to impose a pressure of 0 to 180 (± 10%) to 0 psig in a time period of 60 milliseconds, with a rate of cycling of one cycle every two seconds (see figure 7). Accumulator or similar means shall be provided to vary this pressure to each assembly in alternate operation. The pressurizing system shall also include regulator to insure a constant pressure being delivered to the test assemblies at each cycle.

4.7.9.3 Instrumentation. Each test assembly shall have its pressure transducer connected to a pressure recorder equipped with a galvanometer and electronic counter to record the cycling pressure of each assembly.

4.7.9.4 Test procedure. Coupling assemblies shall be pressurized by starting the pump. The accumulator gate valve shall be adjusted to control the pressure and rise time in conjunction with the pump pressure regulator and the accumulator nitrogen charge (see figure 6). The timer on the electrically operated multiport valve shall be adjusted to produce a pressure surge of 0 to 180 psig to 0 psig in 2 second intervals with the pressure surge peaking to be measured in seconds or fractions of a second. The assemblies shall be subjected to 50,000 cycles at room temperature. There shall be no evidence of leakage or malfunction from any coupling during the cycle surge pressure test.

4.7.9.5 Disassembly and inspection. Upon completion of all tests, the couplings shall be disassembled and each part and tubing end examined for defects.

4.7.10 Burst pressure,

4.7.10.1 Burst pressure test at temperature extremes. The test coupling assembly shall be subjected to a pressure of 375 ± 5 psig, with the test fluid at a temperature of -65°F for 5 minutes. The burst pressure test shall then be repeated using test fluid at a temperature of 200 °F and a pressure of 375 ± 5 psig for 5 minutes. The test fluid shall be in accordance with MIL-PRF-7024 for both tests.

4.7.10.2 Burst pressure test at room temperature. The test coupling assembly shall be subjected to a pressure of 375 ± 5 psig for 5 minutes with the test fluid at a temperature of 75 °F. The test fluid shall be in accordance with MIL-PRF-7024.

4.7.10.3 Disassembly and inspection. Upon completion of the burst pressure tests, the coupling shall be disassembled and each part and tubing end examined. Any evidence of cracks, ruptures, or any other structural deformation or failures of any coupling part shall be cause for rejection of the test coupling.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The flexible couplings are intended for use in aircraft power plant fuel systems to connect variable cavity ferrule tubing end outside the aircraft fire zones where operating temperatures and pressures are within the limits specified in this specification.

6.1.1 Military application. This specification is maintained to support a military aircraft fuel system which has swaged ferrule tubing end with variable cavity and is available for use in other aircraft with similar tubing design. Swaged ferrule tubing with variable cavity has no known commercial application.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Part number (PIN) of coupling required (see 1.2.1).

c. Name and address of the first article inspection laboratory and the name of the Government activity responsible for conducting the first article inspection program.

d. Packaging and packing (see 5.1) including cleaning and marking requirements (see 6.4 and 6.5).

6.3 First article data. When first article samples are submitted (see 6.2) they should be accompanied by a complete inspection report showing the results of the contractor's inspections. The inspection report should include the following:

(a) Report of all inspections graphically presented when possible, together with a detailed statement indicating compliance or extent of noncompliance with all the requirements of this specification and applicable specification sheet referring specifically to paragraph numbers. Whenever a requirement is considered to be not applicable, the report should so state.

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(b) Diagrams of all inspection setup and a complete description of inspection equipment and inspection procedure.

(c) Reproducible outline and description inspection conditions. Where inspections in the specification are substituted, the reason for the substitute inspection should be clearly described.

(d) Copies of log sheets and photographs if available.

6.4 Cleaning and drying. Internal and external surfaces of the coupling and coupling parts must be cleaned and thoroughly dried before delivery.

6.5 Marking. In addition to any special or other identification marking required by the contract, each container should be marked in accordance with MIL-STD-129 and unless otherwise specified should include bar code markings. Marking for commercial packages and interior and exterior packs should be in accordance with ASTM-D3951. The complete military and contractor's or part number as applicable including CAGE (Commercial and Government Entity) Code should be marked on all units.

6.6 Subject term (key word) listing.

Adapter Fluid
Aircraft System
Connector Tubing
Variable Cavity Ferrule Tubing Joint
Vehicle Sub Systems

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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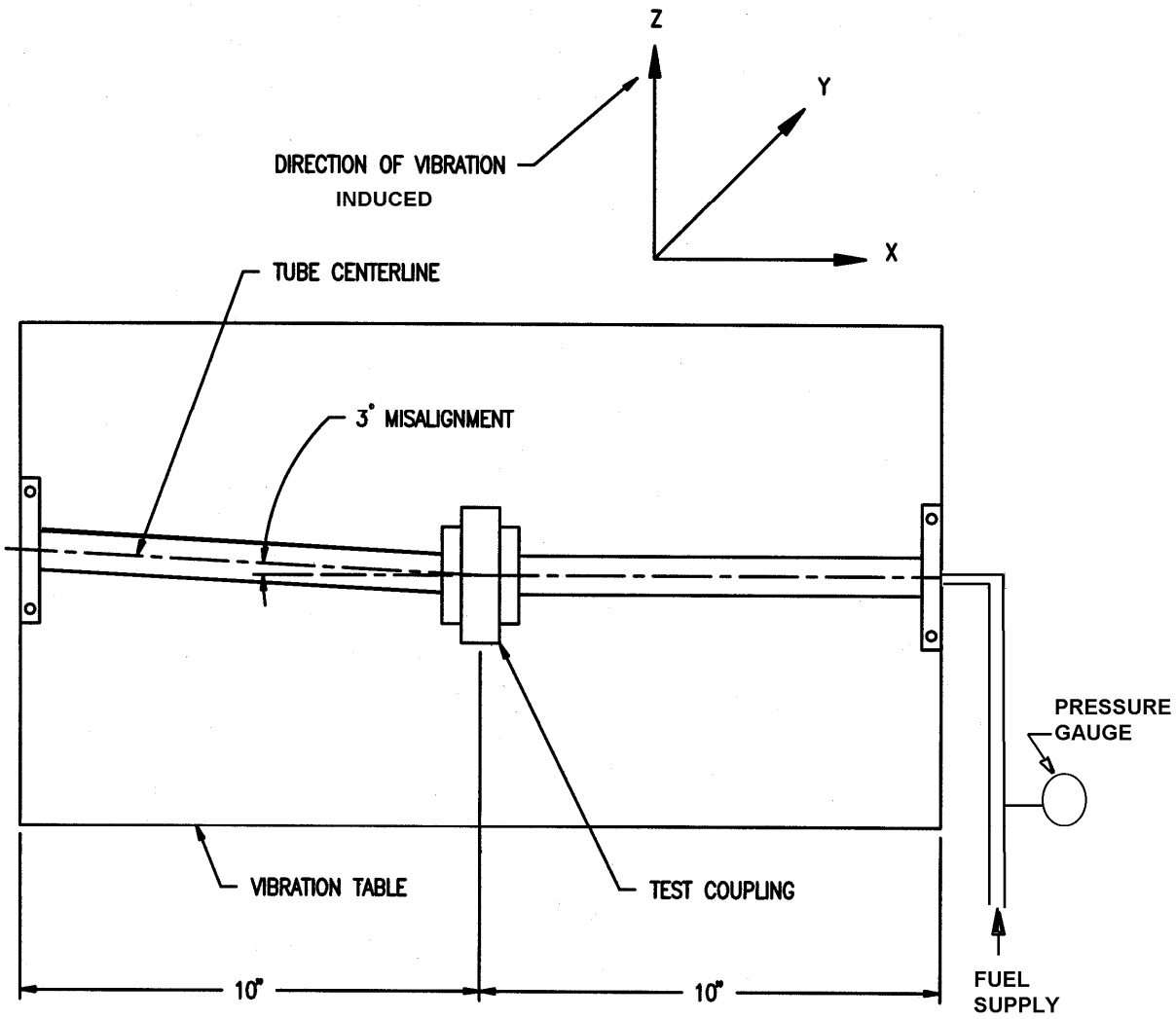


FIGURE 3. Vibration test setup.

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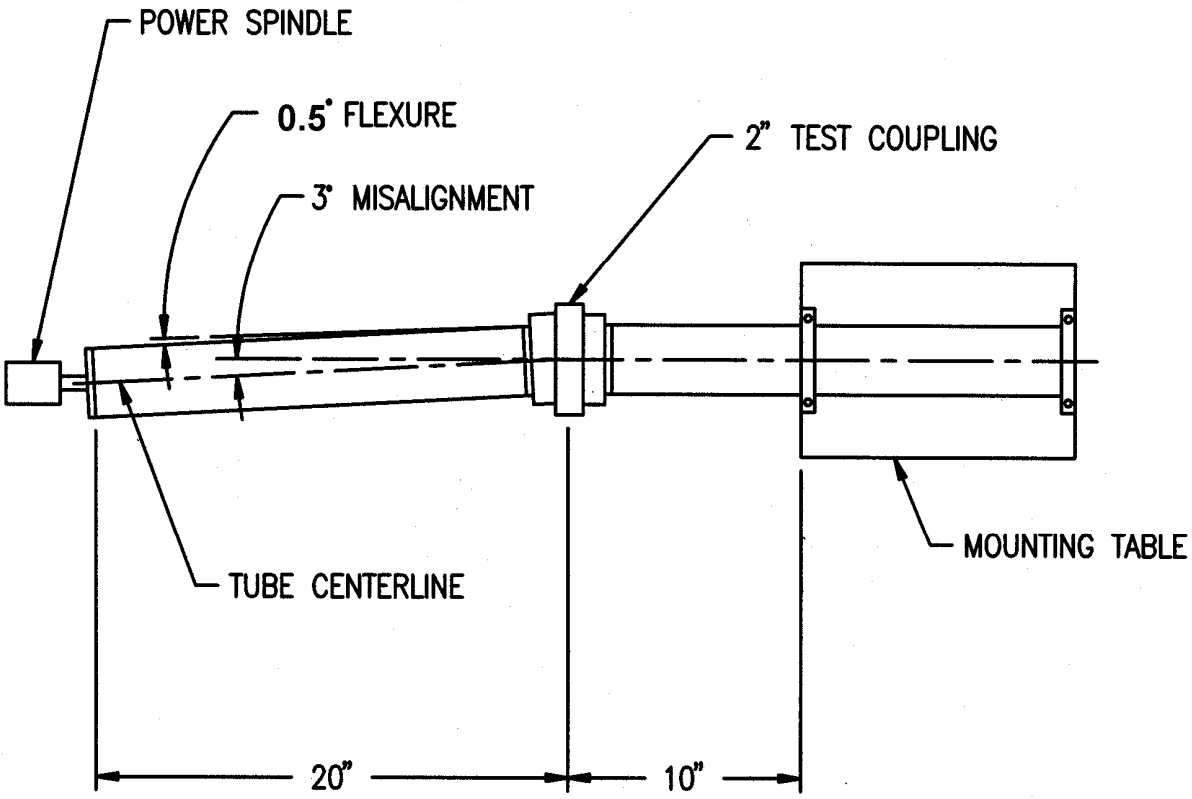


FIGURE 4. Flexure test setup.

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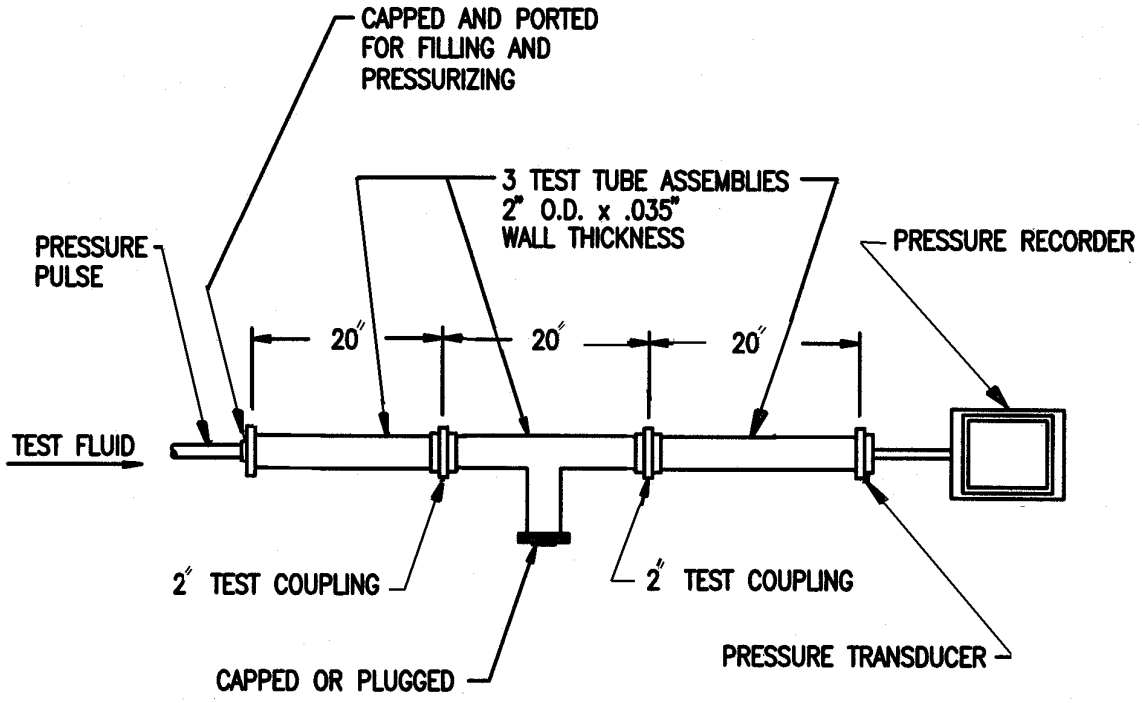


FIGURE 5. Surge pressure test systems.

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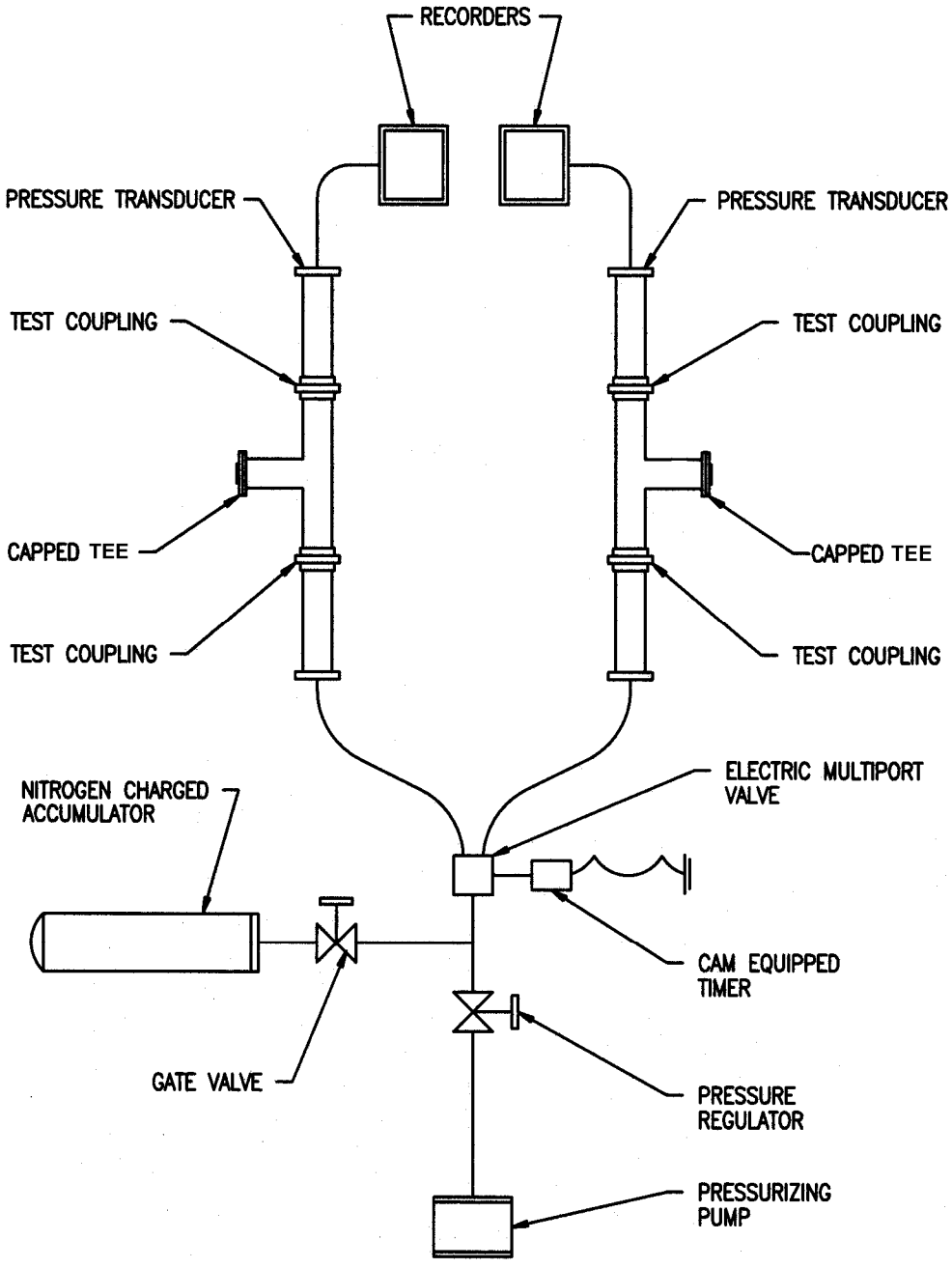


FIGURE 6. Surge pressure test system.

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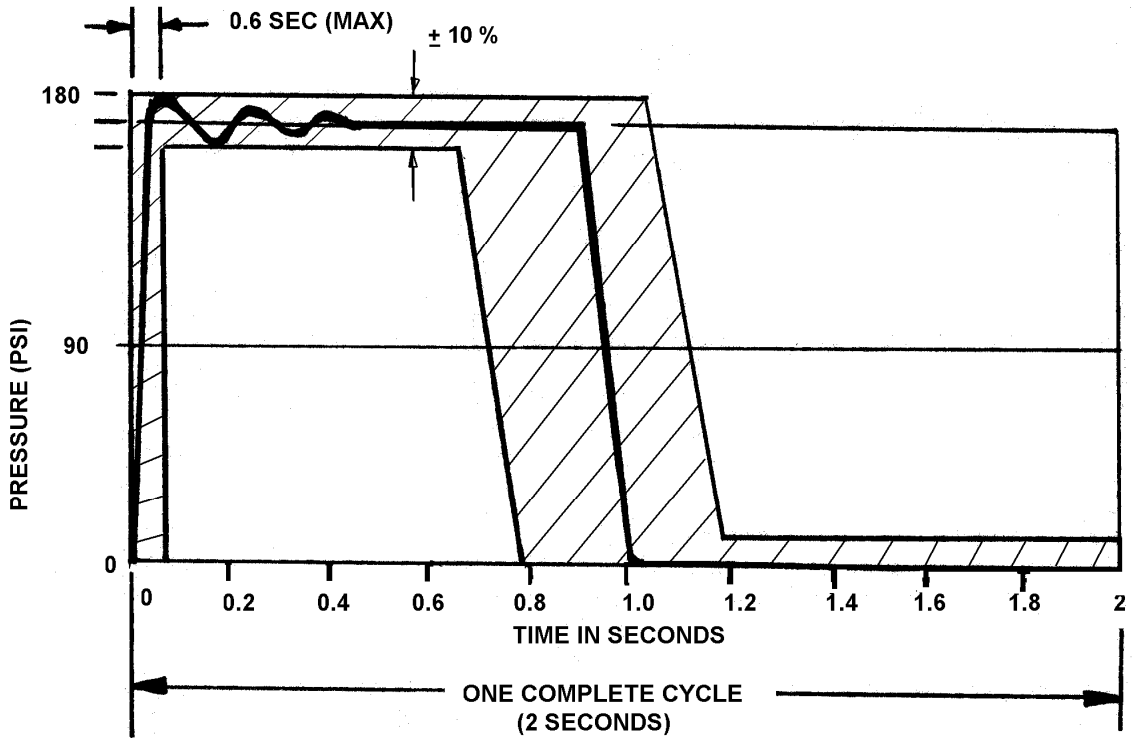


FIGURE 7. Surge pressure curve.

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CONCLUDING MATERIAL

Custodians:
Army-AV
Navy-AS
Air Force-11

Review activities:
Army-AT, MI
Navy-SA, SH
Air Force-71, 99
DLA-CC

Preparing activity:

Navy-AS
(Project 4730-2009-093)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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